

Qualifying Exam in Microeconomics
Arizona State University
August 2022

There are four questions and you have four hours to complete the exam. If you believe that a question is in error, please explain the error and continue working.

1. An exchange economy has $L \geq 2$ goods and $N \geq 1$ consumers. Let $z : S \rightarrow \mathbb{R}^L$ be the *excess demand function* for this economy, where $S = \{p \in \mathbb{R}_{++}^L \mid \sum_{\ell} p_{\ell} = 1\}$, the set of normalized, *positive* prices. Consider these conditions on the excess demand function z , that it

(I) is continuous on S ; and

(II) satisfies Walras's Law: for every $p \in S$, $p \cdot z(p) = 0$.

Consider too this additional property, the *Weak Axiom of Revealed Preference* (WARP):

For any p, p' in S with $z(p) \neq z(p')$, if $p' \cdot z(p) \leq 0$, then $p \cdot z(p') > 0$.

REMARK: Not all authors define WARP for excess demands this way. Use this definition.

- (a) Suppose for this part only that the economy has one consumer; that the consumer's endowment is $\omega \gg 0$; and that the consumer's preferences are represented by a *continuous*, *strictly increasing*, and *strictly quasiconcave* function u .
- i. Does the excess demand function for this economy satisfy (I) and (II)? (You may simply refer to facts we used in the first-year sequence.)
 - ii. Does it follow that the excess demand satisfies WARP? Provide a proof if it does or a (possibly-graphical) counterexample if it does not follow.
- (b) Return to the case of $N \geq 1$ consumers. Prove that if z satisfies (I), (II), and WARP, then the set of Walrasian equilibrium (WE) prices is *convex*. (A point $p^* \in S$ is a WE price if $z(p^*) = 0$.)

2. An economy has one physical good and two states of the world. Each consumer has consumption set $X = \mathbb{R}_+^2$ and endowment $\omega = (\omega_1, \omega_2) \gg 0$, where ω_s is the consumer's endowment of the good in state s for $s = 1, 2$. The price of the good in state s is $p_s > 0$. Each consumer's budget constraint is $p_1 c_1 + p_2 c_2 \leq p_1 \omega_1 + p_2 \omega_2 = w$.

There are two possible consumers, $t = 0, 1$. Both consumers have subjective-expected-utility preferences with belief $\pi = (\pi_1, \pi_2) \gg 0$ about the states, and they have the same endowment. They only differ in their von Neumann - Morgenstern (vN-M) utilities. Consumer t 's vN-M utility is u_t , a continuously differentiable function with $u'_t > 0$. (Neither is necessarily concave.)

In addition, for every $z \geq 0$, $u_1(z) = T(u_0(z))$, where $u = u_0$ and $T : \mathbb{R} \rightarrow \mathbb{R}$ is a *strictly concave function*. In words, consumer 1 is *strictly more risk averse* than consumer 0.

Let c^t solve consumer t 's problem, and, if necessary, relabel the states so that $c_1^0 \leq c_2^0$.

Prove that $c_1^1 \geq c_1^0$ in two steps.

- (a) First suppose that $c_1^0 = c_2^0$; prove that $c_1^1 \geq c_1^0$.
- (b) Next suppose that $c_1^0 < c_2^0$; prove that $c_1^1 \geq c_1^0$. (Hint: consider, as an preliminary step, what consumer $t = 1$ would chose if it were constrained to choose $c_1 \leq c_1^0$.)

You might find this notation useful:

$$\bar{c} = \frac{w}{p_1 + p_2};$$

it is the just-affordable consumption in each state when consumption in each state is equal.

3. Consider the following version of the signaling model of the labor market. There is a worker with ability $\theta \in \{\theta_\ell, \theta_h\}$, $0 < \theta_\ell < \theta_h < 1$, and two firms. Before entering the labor market, a worker with ability θ can choose a training level $\tau \geq 0$, which has a utility cost $c(\tau, \theta) = (1 - \theta)(e^\tau - 1)$ (e is the Euler's number). If a worker with ability θ gets hired at wage w and obtains training τ , then the worker's payoff is $w - (1 - \theta)(e^\tau - 1)$ while the firm's payoff is $\theta - w$. The probability that the worker is of high ability is $p \in (0, 1)$. The timing is as follows: nature draws the worker's θ (that is, draws $\theta = \theta_\ell$ with probability $1 - p$ and $\theta = \theta_h$ with p), which is privately observed by the worker; the worker chooses a level of training $\tau \geq 0$; the two firms observe the worker's training chosen but not the worker's ability and then compete à la Bertrand by each posting a wage offer; finally, the worker chooses one of the two firms (or none of them), breaking ties equiprobably. The equilibrium concept is Weak Perfect Bayesian Equilibrium (WPBE) in pure strategies (except for the tie-breaking rule) with the added condition that firms have "common beliefs" off and on the equilibrium path.
- Define a WPBE in this setting. (Hint: Feel free to ignore the last stage since the worker either rejects both firms or goes to the one with the higher wage or randomizes between the two in case of a tie.)
 - Show that in any separating WPBE, each firm pays a wage equal to θ_h if it observes the equilibrium training level that the high-ability worker chooses, and pays a wage θ_ℓ if it observes the equilibrium level of training that the low-ability worker chooses. Show also that in a separating WPBE, the training level of the low-ability worker is $\tau = 0$.
 - Derive the interval of training levels that can be part of a separating WPBE. Derive a least-costly separating WPBE. Accompany your derivations with pictures.
 - Explain how with an equilibrium refinement one can eliminate all the separating WPBE except for those involving the least-costly separating training level. If you have time, explain also how the pooling WPBE can be eliminated as well.

4. Copper and zinc are used in fixed proportions to make brass. Normalize units so that one unit of zinc and one unit of copper make one unit of brass. There are two firms: one produces copper and the other produces zinc. Each firm has zero cost (or all cost is fixed and sunk).

The demand for the final good *brass* is given by a function $D : \mathbb{R}_+ \rightarrow \mathbb{R}_+$, where D satisfies the usual properties:

- D is *continuous* and *nonincreasing* on $\mathbb{R}_+$;
- on $\{P \in \mathbb{R}_+ \mid D(P) > 0\}$, it is *continuously differentiable* with $D'(P) < 0$, where P is the price a consumer must pay for a unit of brass; and in addition,
- D is *twice differentiable* and *concave* on $\{P \in \mathbb{R}_+ \mid D(P) > 0\}$.

Define $\bar{P} = \sup\{P \in \mathbb{R}_+ \mid D(P) > 0\}$. Finally also assume that

- $0 < \bar{P} < \infty$; that $D(0) < \infty$; and $D'(\bar{P}) < 0$. (The derivative is from the left.)

The firms compete in prices. Specifically, the firms independently and simultaneously choose prices; the resulting price for a unit of the final good *brass* is then $P = p_1 + p_2$, the sum of the prices of copper and zinc.

Set each firm's strategy set to $[0, \bar{P}]$. Each firm's preferences over price profiles are represented by profit: $u_i(p_1, p_2) = D(P)p_i$ for $i = 1, 2$.

- (a) As written, is this a *supermodular* game (aka a *game with increasing differences* or a *game with strategic complementarities*)? Explain.
- (b) Define a function $g : [0, \bar{P}] \rightarrow [0, \bar{P}]$ to be the solution x to the equality

$$D'(P)x + D(P) = 0 \tag{1}$$

if the solution x is nonnegative; if the solution x is negative, then set $g(P) = 0$; and set $g(\bar{P}) = 0$. Is g continuous? Is it monotone (increasing or decreasing)? In each case, explain.

- (c) Define a function h on $[0, \bar{P}]$ by

$$h(P) = g(P) + g(P) - P. \tag{2}$$

- i. What is the significance of the equality $h(P) = 0$ from the viewpoint of a Nash equilibrium (NE) of this game?
- ii. Does it follow that a NE exists? If it does, then explain why. If it doesn't follow, then what additional assumptions on g would ensure that one exists?
- iii. If a NE *does* exist, then, among NE with *positive production* ($D(P) > 0$), is it *unique*? Explain. (A picture might prove helpful.)